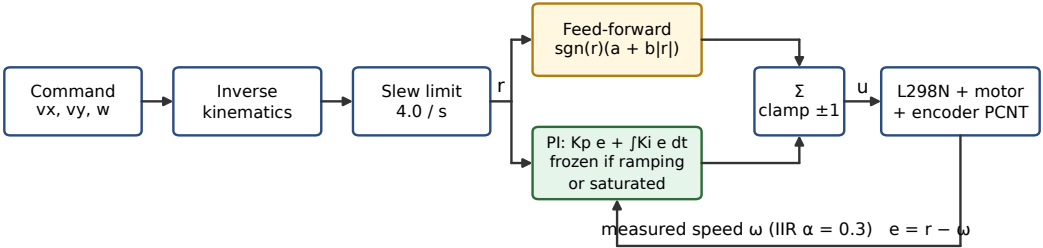
Command
vx, vy, w
Inverse
kinematics
Slew limit
4.0 / s
r
Feed-forward
sgn(r)(a + b|r|)
PI: Kp e + ∫Ki e dt
frozen if ramping
or saturated
Σ
clamp ±1
u
L298N + motor
+ encoder PCNT
measured speed ω (IIR α = 0.3)
e = r − ω